

- Without further assumptions, given a single realization of time series  $\{x_1, x_2, \dots, x_n\}$ , it is almost impossible to estimate mean function  $\mu(t)$  & cov. function  $\gamma(s, t)$ .

- One such type of assumption: weak stationarity.

$$\mu(t) = \text{const.}$$

$\gamma(t+h, t)$  depends only on  $h$ .

- Sample mean under weak stationarity.

$$\bar{X}_n = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Assume  $\{x_t\}$  i.i.d.  $\text{Var}(x_t) = \sigma^2$

$$\begin{aligned} \text{Var}(\bar{X}_n) &= \text{Var}\left(\frac{1}{n}(x_1 + x_2 + \dots + x_n)\right) \\ &= \frac{1}{n^2} \text{Var}(x_1 + x_2 + \dots + x_n) \\ &= \frac{1}{n^2} [\text{Var}(x_1) + \dots + \text{Var}(x_n)] \end{aligned}$$

$$= \frac{\sigma^2}{n}$$

↳ foundation for constructing confidence interval & carrying out hypothesis test about  $\mu = E[X_t]$ .

E.g. 95% confidence interval.  
(large sample)

$$\bar{X}_n \pm \frac{1.96}{\sqrt{n}} \hat{\sigma}$$

estimated  $\sigma$ .

- $$E[\bar{X}_n] = E\left[\frac{1}{n} (X_1 + X_2 + \dots + X_n)\right]$$

$$= \frac{1}{n} \{ E[X_1] + \dots + E[X_n] \}$$

$$= \mu$$

unbiased

Question: how about weakly stationary time series  $\{X_t\}$

- $$E[\bar{X}_n] = \dots \text{(same calculation as above)}$$

$$= \mu$$

$$\bullet \text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n}(X_1 + \dots + X_n)\right)$$

$$= \frac{1}{n^2} \text{Var}(X_1 + \dots + X_n)$$

$$\text{Var}(X_1 + \dots + X_n) = \text{Cov}(X_1 + \dots + X_n, X_1 + \dots + X_n)$$

$$= \sum_{s=1}^n \sum_{t=1}^n \text{Cov}(X_s, X_t)$$

$\gamma(s-t)$

$$= \sum_{s=1}^n \sum_{t=1}^n \gamma(s-t)$$

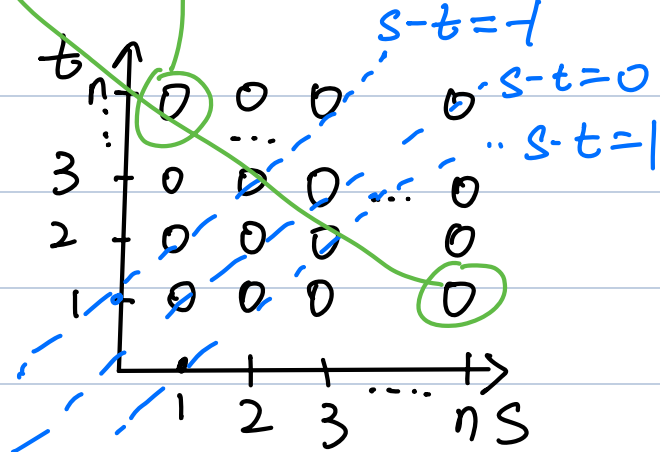
Q: How many  $(s, t)$  pairs correspond to  $s-t=0$

A: 1

Q: ...  $(s, t)$  pairs ...  $|s-t|=1$

A:  $n-1$

$$= n\gamma(0) + (n-1)(\gamma(1) + \gamma(-1)) + (n-2)(\gamma(2) + \gamma(-2)) + \dots + (\gamma(n-1) + \gamma(1-n))$$



$$= \sum_{h=-(n-1)}^{n-1} (n - |h|) \gamma(h)$$

$$\hookrightarrow \text{Var}(\bar{X}_n) = \frac{1}{n^2} \sum_{h=-(n-1)}^{n-1} (n - |h|) \gamma(h)$$

$$= \frac{1}{n} \sum_{h=-(n-1)}^{n-1} \left(1 - \frac{|h|}{n}\right) \gamma(h)$$

Recall for i.i.d case

$$\text{Var}(\bar{X}_n) = \frac{1}{n} \sigma^2 = \frac{1}{n} \gamma(0)$$

reduces to when  
 $\gamma(h) = 0$   
 for  $h \neq 0$